AI for Medical Imaging

Master Artificial Intelligence

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Fall 2026

Universiteit van Amsterdam
Instituut voor Informatica

Amsterdam UMC

2026/09/01 – Introduction lecture

Who am I

Dr. Hoel Kervadec, assistant prof. in the qurAI group at UvA

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Engineering degree in France, PhD in Canada, post-doc in the Netherlands

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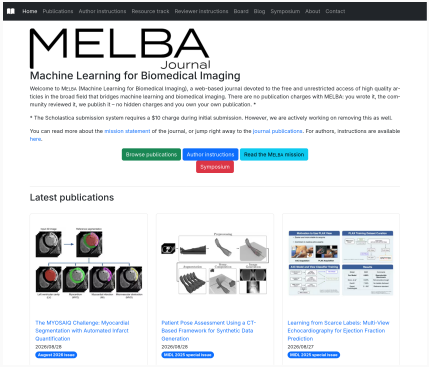
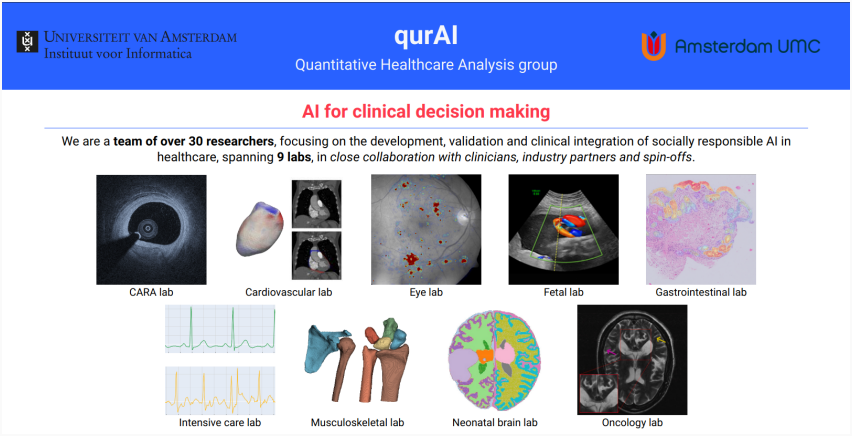
Board member of the MIDL Foundation: https://midl.io



Publishing editor of the MELBA journal: https://melba-journal.org

Quantitative healthcare analysis (qurAI) group

Inter faculty group between *IvI* and the *Amsterdam UMC*



https://qurai.amsterdam/

2026/09/01 – Introduction lecture

Medical imaging? with AI?

What is medical imaging?

Keyword is: modalities ; there are many of them

Most of them non-invasive, used for diagnostic and (surgery) planning

Ultrasounds - I

Ultrasound acquisition

Embryo at 14 weeks

Ultrasounds - II

A modern US scanner

Ultrasounds - II

A modern US scanner

but now portable exist

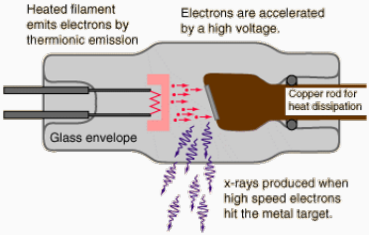
X-Rays - I

Discovered by Wilhelm Röntgen

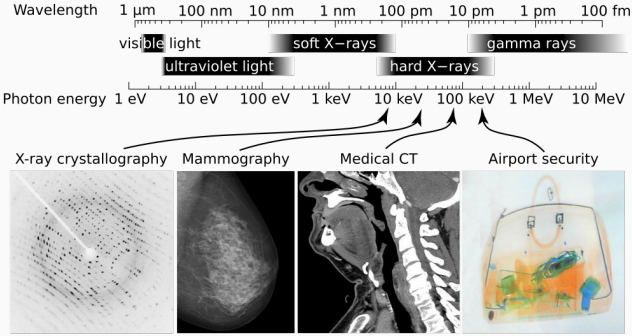
Hand mit Ringen (1895)

Frontal view of a knee

X-Rays - II



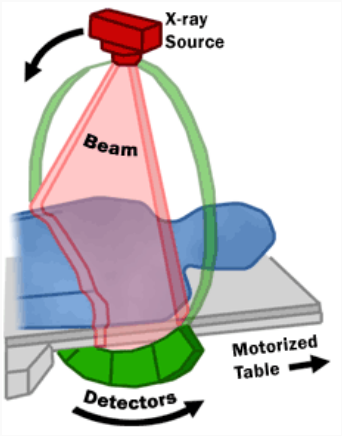
X-ray production



Different applications of x-rays

Computer Tomography (CT) - I

"Logical" next step: rotating X-ray for 3D view

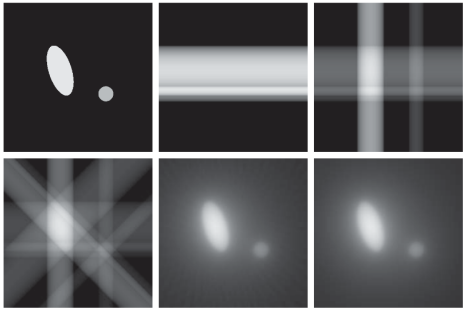


An x-ray beam rotates around the patient

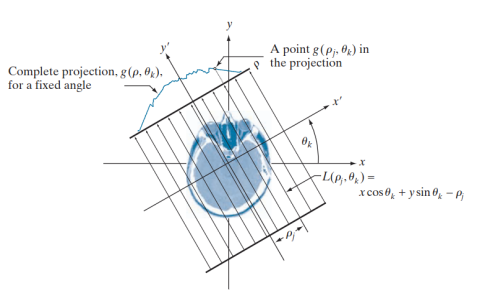
Some modern CT

Computer Tomography (CT) - II

Tomographic reconstruction



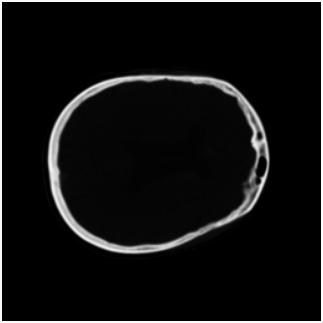
Reconstruction for various numbers of projections



Projection for a specific angle

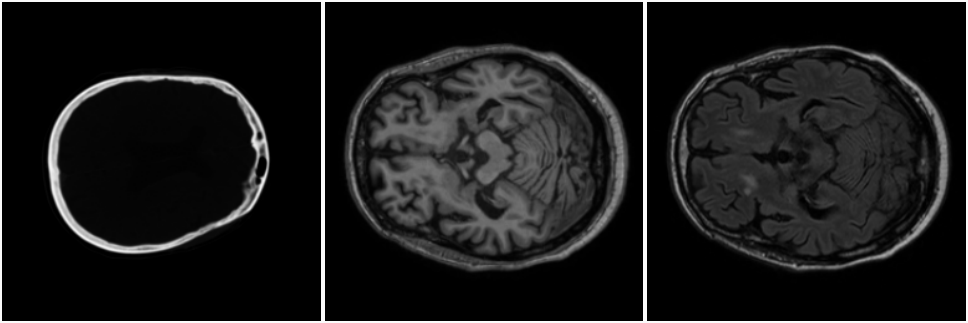
Source: *Digital image processing, 4th edition, Gonzalez and Woods, Section 5.11*

MRI - I



CT

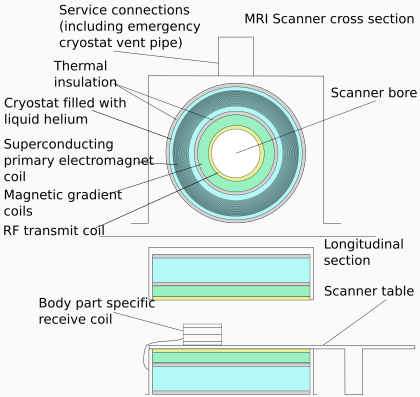
MRI - I



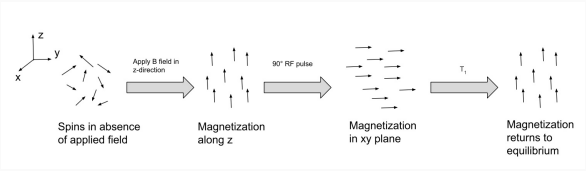
CT MRI (T1) MRI (Flair)

Images of a brain

MRI - II



Cross section of an MRI machine



Spin orientation during relaxation

MRI - III

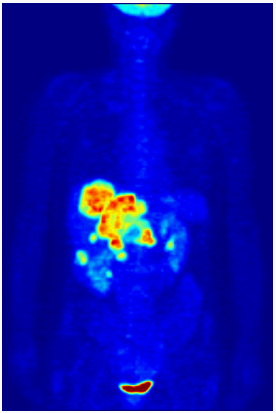
Nieuwe MRI-scanners worden EMC binnengetakeld

30 oktober 2017, 11:58 • Aangepast 20 december 2021, 19:20 • 1 minuut leestijd

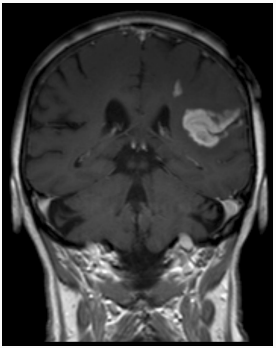
https://www.rijnmond.nl/nieuws/160822/nieuwe-mri-scanners-worden-emc-binnengetakeld

Many others

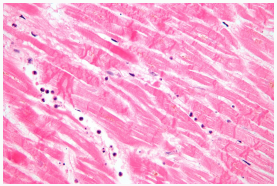
PET scan, Photon-counting CT, scanners with contrast agent, ...



PET scan



MRI with contrast



Histopathology

Trade-offs - I

Cost, capabilities, compatibilities, patient comfort

Modality	Physical agent	Strength	Weakness	Comment
Plain film	X-ray	Global view of anatomy	Does not reveal fine detail of organs	Least expensive
Ultrasound	Sound waves	Good organ detail, anatomical	Operator-dependent; air and/or bone may prevent visualization of organs	No radiation, no sedation; less invasive; relatively cheap
CT	X-ray	Good organ detail, quick, anatomical	May need sedation and/or contrast	Moderately expensive, radiation
MR	Electromagnetic waves	Great tissue contrast, multiple planes; anatomical; new inroads to functional imaging	Relatively slow, needs sedation in younger children; may need contrast	Most expensive; no radiation
Nuclear	Gamma rays	Physiological (functional)	Organ-specific; poor on anatomic detail	Moderately expensive

Source: *Huda W, Slone RM, Aaland MT (2002)*

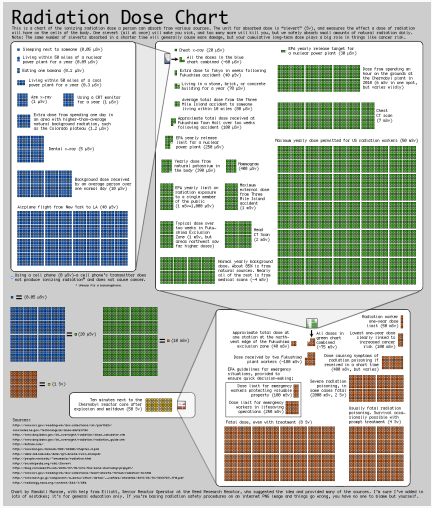
Trade-offs - II

Cost, capabilities, compatibilities, patient comfort

TABLE 2–1. Typical upper-end characteristics and capabilities of some of the principal imaging modalities. Entries for a particular imaging system or study may be quite different. See also Table 21–1. Note that costs are very study-specific, and that the figures presented here do not include physician fees.

X	Probe/Signal Detected	Signal Detector	Tissue-Attribute Imaged	Information Primarily on Anatomy (a), or Physiology (p)	Soft Tissue Contrast (1 = best)	Spatial Resolution (mm)	Imaging Time (min)	Cost, 2004 ($K)
X-ray	X-rays transmitted through body	Screen/film	Density, effective atomic number, thickness	a	3	<0.1	10+	0.2
Mammo		Screen/film		a	3	<0.1	10+	0.2
CR		PSP		a	3	<0.3	10+	0.2
DR		FPD		a	3	<0.3	10+	0.2
Mammo		FPD (digital)		a	3	<0.2	10+	0.2
Fluoro		II & TV/CCD		a	3	<0.3	30+	0.4
DSA		FPD, or II & CCD		a	1	<0.3	10+	2.5
CT		X-ray detector array		a	2	<0.5	10+	1.2
NM	Gamma rays emitted from organ	NaI crystal	Radionuclide uptake by organ	p	1	<3.0	30+	1.0
SPECT		NaI crystal		p	1	<3.0	30+	1.0
PET		Gamma ray detector array		p	1	<3.0	40+	1.5
MRI	r.f. absorbed, re-emitted	r.f. coil	Proton spin relaxation	a, p	1	10	40+	2.0

Trade-offs - III



https://xkcd.com/radiation/

Why AI?

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More and more images to process

Resolution increase, acquisition time goes down

New modalities

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Anomaly detection, triage, reconstruction and visualisation, ...

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Quantification vs decision

Help support clinicians, not replace them

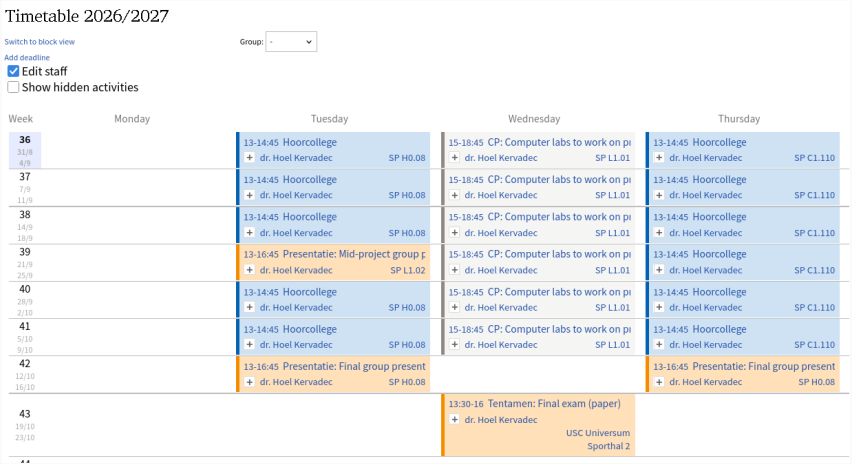
2026/09/01 – Introduction lecture

Break

2026/09/01 – Introduction lecture

Course structure

Schedule - I



Current Datanose schedule

Schedule - II

12 lectures in total (some with guests), 6 laptop sessions for the project

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Mid-project presentation, final project presentation, exam

Grading - I

- Group project (40%)
  - Mid-project presentation (10%)
  - Final project presentation (20%)
  - Deliverable (10%)
- Exam on paper (60%)

The exam expects you to have been involved in the project.

Grading - II

Presentations: everyone must present

Must attend *all* presentations of your day, both days recommended

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Exam: one a4 sheet of notes, both sides, allowed.

Test your knowledge, and understanding.

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Must attend *all* presentations of your day, both days recommended

Exam: one a4 sheet of notes, both sides, allowed.

Test your knowledge, and understanding.

Recommendation letters: always happy to write them, but I will rely more on the exam results when writing it.

Materials

- Lectures slides;
- Summaries (drawings) of some elements;
- 4 papers to read over the course;

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- Anki deck (flash cards, experimental);
- lecture notes.

Bonus materials

https://www.museumvrolik.nl/

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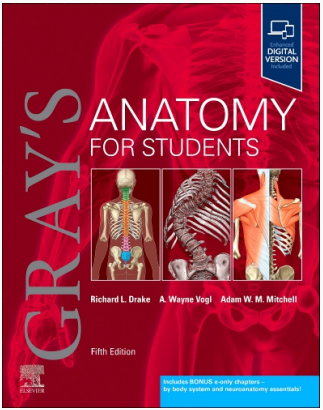

Gray's anatomy

Sobotta atlas of anatomy

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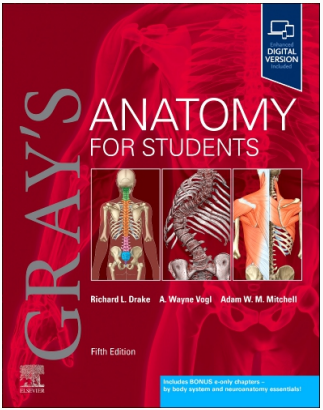

Gray's anatomy

Sobotta atlas of anatomy

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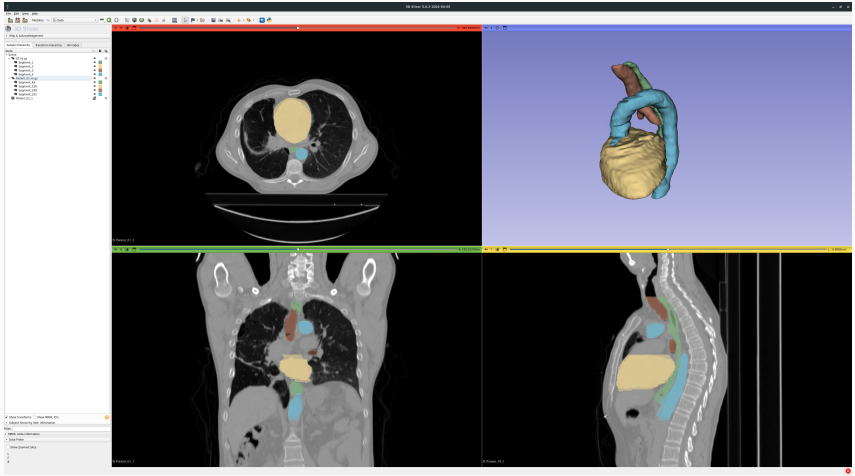
Project

Project overview - I

Group of **6**, image segmentation task. One TA assigned per group.



Project overview - II

Baseline is given (2D CNN), have to:

- make it better;
- quantify how much better it is.

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Bonus weekly assignments (ungraded)

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Bonus weekly assignments (ungraded)

Very open, on purpose. Lectures content is mapped with need for the project.

Teaching assistants to guide, suggest, and advise. **Not to debug.**

Two phases for the project

First half (until mid-project presentation):

- only half of the data available;
- compute budget for the whole course

Second half (until the end):

- all data available;
- compute budget per group.

Deliverables

Mid-project presentation: upload slides in advance

Final presentation: upload slides + results + git repository in advance

Group formation

Self-signup on Canvas, until Thursday 3th at 16:00, then we fill the groups ourselves randomly.

Master AI and non-master AI students: balanced groups.

`https://canvas.uva.nl/courses/59853/groups#tab-44347`

We will rebalance the groups if required.

Groups to be final by Friday 4th at noon.